# MomentNoise Documentation

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October 27, 2026, for MomentNoise 0.2.0

## 1 Introduction

MomentNoise is a noise generator whose amplitude distribution can be changed: besides the level, you set the *skewness* (asymmetry) and the *excess kurtosis* (tail weight) of the noise. The default is Gaussian (normal) noise at −24 dBFS RMS. The lower half of the window shows the theoretical probability density function (pdf) of the noise. Together with its companion plug-in AmplitudeHistogram, which measures the histogram of a signal, it shows how the statistics of a signal and its histogram belong together.

**Warning:** heavy-tailed noise (large excess kurtosis) has rare peaks far above its RMS level. At −24 dBFS RMS and excess kurtosis 100, single peaks within 10 seconds reach +7 to +15 dBFS (about 35 dB above the RMS level); at excess kurtosis 15 about +5 dBFS. Lower the level before you listen to such settings. Very large excess kurtosis sounds like crackles: rare strong clicks in quiet noise.

MomentNoise is a generator: the input signal is replaced by the noise (independent noise in every channel).

## 2 Installation

The downloaded file is a zip file. It contains the plug-in (.vst3; on macOS also the AU, .component), the Standalone application, a ReadMeFirst.txt, the licence files and this manual. Copy the plug-in to the right place and let your DAW rescan its plug-ins.

```
Windows (VST3):  C:\Program Files\Common Files\VST3
Mac (VST3):      /Users/yourUSERNAME/Library/Audio/Plug-Ins/VST3
Mac (AU):        /Users/yourUSERNAME/Library/Audio/Plug-Ins/Components
Linux (VST3):    /home/yourUSERNAME/.vst3
```

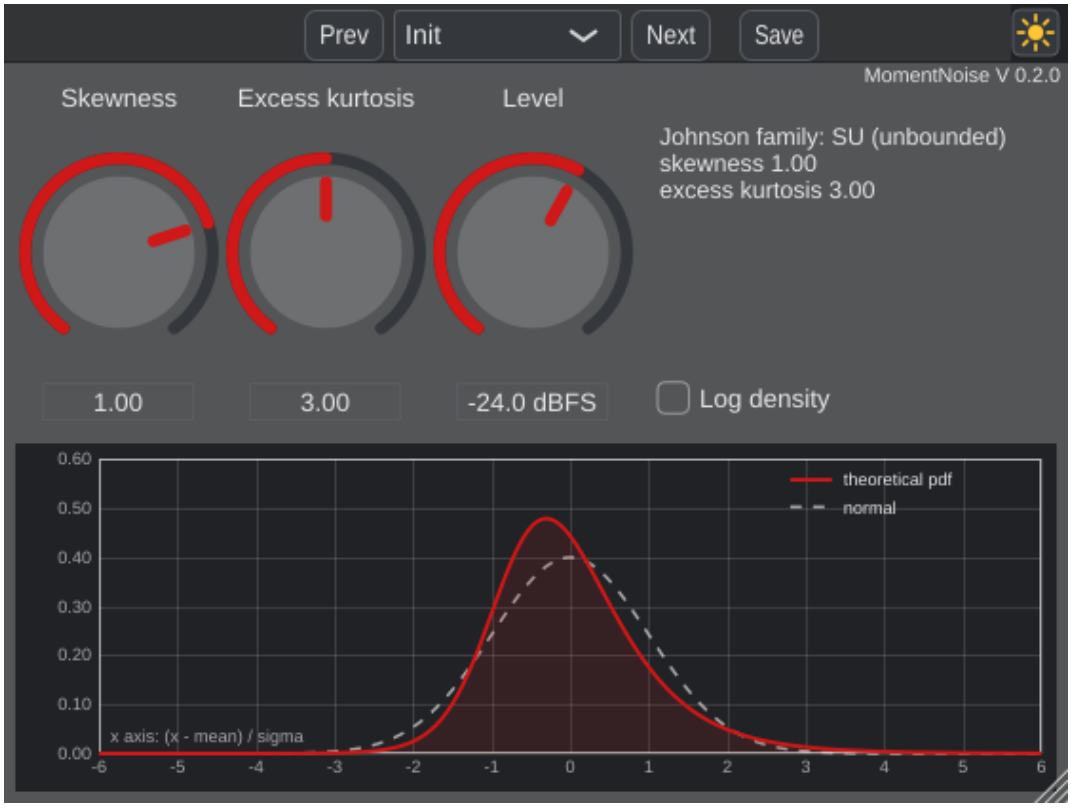
The macOS version is not signed with an Apple Developer ID. If macOS reports that the plug-in is damaged or cannot be opened, remove the quarantine flag in the Terminal, e.g. `xattr -cr ~/Library/Audio/Plug-Ins/VST3/MomentNoise.vst3`.

### 2.1 Presets and settings

The presets (plain XML files) and the settings file (`user.settings`: day/night theme) are stored in this folder – delete it to remove MomentNoise completely:

```
Windows: C:\Users\yourUSERNAME\AppData\Roaming\Jade\_Hochschule\MomentNoise
Mac:     /Users/yourUSERNAME/Library/Audio/Presets/Jade\_Hochschule/MomentNoise
Linux:   /home/yourUSERNAME/.config/Jade\_Hochschule/MomentNoise
```

# 3 How to use




1. **Preset handler**: Prev/Next step through the presets. To save a new preset, type a name and press Save. Save turns red when you change a parameter of the displayed preset; press it to update the preset. To delete a preset, delete its XML file.

2. **Day/night**: switches between the night (dark) and the day (light) theme.

3. **Skewness** $(-2 \dots 2)$: the 3rd standardized moment. 0 is symmetric; positive values give a long tail towards positive amplitudes, negative values towards negative amplitudes.

4. **Excess kurtosis** $(-1.5 \dots 100)$: the 4th standardized moment minus 3. 0 is Gaussian; negative values give light tails (uniform noise: $-1.2$), positive values heavy tails with rare large peaks (Laplace: 3). Every distribution has an excess kurtosis above skewness$^2 - 2$; MomentNoise keeps a distance of 0.5 to this bound and raises smaller values (the info text says so). The knob is exponential, with 3 in its middle: 0 is at about one third of the turn, 10 at two thirds, 25 at three quarters, so the small values keep most of the knob. For a precise value, type it into the text box. Above about 50, a few seconds of the noise do not show the set value any more: the measured kurtosis depends on a few very rare, very large samples (AmplitudeHistogram then shows less).

5. **Level**: RMS level (standard deviation) of the noise in dBFS.

6. **Info text**: the Johnson family used (see section 5) and the skewness and excess kurtosis that were reached.

7. **Log density**: the density axis of the display logarithmic (5 decades) or linear. The tails, where the kurtosis shows, are only visible in the logarithmic view.

8. **Display**: the theoretical pdf (red) over the standardized amplitude $(x - \text{mean})/\sigma$ from $-6\sigma$ to $6\sigma$, the normal pdf as dashed reference. AmplitudeHistogram uses the same axes, so both displays can be compared directly.

# 4 Controls

All controls, their range and default value. Rest the mouse on a control to see its description as a tooltip.

| Control | Range | Default | Description |
|---|---|---|---|
| Skewness | -2.00 - 2.00 | 0.00 | Asymmetry of the amplitude distribution (3rd standardized moment): 0 = symmetric, positive = long tail towards positive amplitudes. |
| Excess kurtosis | -1.50 - 100.00 | 0.00 | Tail weight (4th standardized moment minus 3): 0 = Gaussian, negative = light tails (uniform: -1.2), positive = heavy tails, rare large peaks (Laplace: 3). At least skewness squared - 1.5, smaller values are raised. |
| Level | -60.0 dBFS - 0.0 dBFS | -24.0 dBFS | RMS level of the noise (its standard deviation). Heavy tails give peaks far above the RMS level: they may exceed 0 dBFS. |
| Log density | Off - On | Off | Density axis of the display logarithmic (shows the tails) or linear. |

# 5 How it works

The noise comes from the *Johnson system* of distributions (N. L. Johnson, 1949). A standard normal random variable $z$ is transformed into

$$x = \xi + \lambda\, y(z)$$

with one of four families:

| | | |
|---|---|---|
| SN (normal) | $y = z$ | skewness 0, excess kurtosis 0 |
| SL (lognormal) | $y = e^{z/\delta}$ | on the "lognormal line" |
| SU (unbounded) | $y = \sinh(z/\delta) + t\cosh(z/\delta)$ | above the lognormal line (heavy tails) |
| SB (bounded) | $y = 1/(e^{-z/\delta} + c)$ | below the lognormal line (light tails, bounded) |

($t = \tanh(\gamma/\delta)$ and $c = e^{-\gamma/\delta}$ are Johnson's parameter $\gamma$ in a form that cannot overflow.) Together the families cover every possible pair of skewness and excess kurtosis, so both can be set

independently. When a parameter changes, a background thread searches $\delta$ and $t$ (or $c$) so that the moments of $y$, computed by numerical integration over $z$, have the wanted skewness and excess kurtosis (two nested root searches, at most about 80 ms). $\xi$ and $\lambda$ then give mean 0 and variance 1; a negative skewness mirrors the distribution. The pdf of the display follows from the change of variables, $p(x) = \varphi(z)\,|dz/dx|$.

For every sample a Gaussian random number $z$ is drawn and transformed; the level scales the result (smoothed over 20 ms).

# 6 Source code

MomentNoise is open source. You find the source code here:
https://github.com/JoergBitzer/noise_moments. Please report bugs there, too.

# 7 Legal stuff

For this documentation the CC-BY 4.0 licence is valid. Copyright holder is Joerg Bitzer. The source code of the plug-in is under the MIT licence. The plug-in binaries contain JUCE (used under the AGPLv3), the VST3 SDK (MIT) and, on macOS, the Audio Unit SDK (Apache 2.0); the binaries as a whole are distributed under the AGPLv3.

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